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29 April 1999

DETAIL SPECIFICATION

AIR CLEANER ASSEMBLY

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 900 cubic foot per minute (ft^3/min) [425 liter per second (L/s)], fender-mounted, heavy-duty, two-stage, dry-type air cleaner assembly referred to herein as the "assembly" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-595/24084 - Green, Semigloss.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-62309 - Filter Element, Air.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

U.S. ARMY DRAWINGS

11669740 - Air Filter Element.
12325888 - Air Cleaner Assembly - Armored (Clean Air).

(Copies of these drawings are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents those cited in the solicitation or contract (see 6.2).

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ASTM INTERNATIONAL

ASTM C800 - Glass Fiber Blanket Insulation (Aircraft Type) (DoD Adopted).

(Copies are available from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or website: http://www.astm.org.)

SAE INTERNATIONAL

SAE AS478 - Identification Marking Methods (DoD Adopted).

(Copies of this document are available from www.sae.org or SAE Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATIO (ISO)

ISO 5011 - Inlet Air Cleaning Equipment for Internal Combustion Engines and Compressors – Performance Testing

(Copies of these documents are available from www.iso.org or American National Standards Institute, 11 West 42nd Street, New York, N.Y. 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.5.

3.2 Material. Materials shall be uniform and free from imperfections or defects which affect their serviceability (see 4.8.1).

3.2.1 Corrosion protection. All metallic parts shall be made from corrosion resistant steels or treated/ plated with corrosion-resistant materials (see 3.7.2 and 4.8.1).

3.2.2. Asbestos and cadmium usage. Asbestos and cadmium materials shall not be used in any form in any part of the assembly (see 4.8.1).

3.2.3 Radioactive material usage. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries (see 4.8.1).

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3.2.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.8.1).

3.3 Design and construction. The assembly shall be constructed as specified herein and in accordance with Drawing 12325888. The assembly shall incorporate a precleaner, filter element and a vehicle exhaust dust ejection system (VEDES) precleaner scavenging manifold (see 4.8.1 and 6.2).

3.3.1 Filter element. The filter element shall be in accordance with MIL-PRF-62309 and Drawing 11669740 (see 4.8.1).

3.3.2 Weight. The weight of the assembly shall not exceed 600 pounds (lb) (272 kilogram (kg)) (see 4.8.1).

3.4 Operating requirements. Except as otherwise specified herein, requirements shall be met under the following standard conditions:

a.	Temperature:	73 ± 18 degrees Fahrenheit (°F) (23 ± 10 degrees Celsius (°C)).
b.	Humidity:	50 ± 15 percent (%).
c.	Density of air:	0.073 pounds per cubic foot (lb/ft^3) [1.17 kilogram per cubic meter (kg/m^3)] at 80°F (27°C).
d.	Pressure:	30 ± 1 in. of mercury (Hg) (101 ± 3 kilopascals (kPa)).
e.	Dust:	Chemical analysis of dust and particle size distribution shall be in accordance with ISO 5011.
f.	Air flow:	900 ft^3/min (425 L/s).
g.	Dust feed rate:	0.025 ± 0.005 gram per cubic foot (g/ft^3) (0.88 ± 0.18 g/m^3) of actual input airflow.

Actual input airflow shall be defined as airflow plus the scavenge airflow for each of the five 10 minute stepped airflow cycles listed in table I. (NOTE: First step would have an actual input airflow of 970 ft^3/min.)

3.4.1 Initial airflow restriction. The initial airflow restriction of the assembly shall not exceed 9 in. of water (2.24 kPa) (see 4.8.2.1)

3.4.2 Dust capacity. When operating with a stepped airflow in accordance with table I and fed with ISO 5011 coarse dust using the prescribed dust injector as specified in ISO 5011, the time for the assembly to reach load capacity (see 6.6.1) shall be not be less than 80 hours. There shall be no dust tracking on the downstream side of the element. The coarse dust accumulative efficiency shall not be less than 99.9% and the precleaner efficiency shall be not be less than 97% (see 4.8.2.2).

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TABLE I. Stepped air flow.

Time (minute)	Rated flow (%)	Air flow ft^3/min	Air flow (L/s)	Scavenge airflow ft^3/min	Scavenge airflow (L/s)
10	100	900	(425)	70	(33)
10	60	540	(255)	107	(50)
10	20	180	(85)	128	(60)
10	80	720	(340)	92	(43)
10	60	540	(255)	107	(50)
10	40	360	(170)	120	(57)

3.4.3 Dust efficiency. The assembly shall remove ISO 5011 fine dust from air at an initial efficiency of not less than 99.5% with a period of 30 minutes using the prescribed dust injector as specified in ISO 5011. If the air cleaner has an air flow velocity rating of 6 ft/min (183 cm/min) or less, the dust shall be fed for 60 minutes (see 4.8.2.3 and 4.8.2.3.1).

3.4.4 Post service airflow restriction. After one operation to load capacity in accordance with 3.4.2, the assembly air flow restriction shall not have increased by more than 1 in. of water (0.25 kPa) above that measured in 3.4.1 (see 4.8.2.4).

3.4.5 Air leakage resistance. There shall be no more than 2 in. of water (0.5 kPa) drop in 3 minutes when the assembly is subjected to a vacuum of 30 $\pm$ 0.5 in. of water (7.5 $\pm$ 0.1 kPa) (see 4.8.2.5).

3.5 Interface requirements. The assembly shall conform to the interface envelope dimensions and mounting in Drawing 12325888 (see 4.8.3). Each assembly shall accommodate the inputs and interfaces specified in 3.3 through 3.3.2.

3.6 Support and ownership requirements. Each assembly shall possess the following life cycle ownership characteristics.

3.6.1 Identification marking. Identification marking shall be permanent and in accordance with SAE AS478, and shall include the maximum rated air flow, part number, month, and year of manufacture (see 4.8.4.1).

3.6.2 Painting. Unless otherwise specified (see 6.2), the preparation, conditioning, and painting of the assembly shall be in accordance with the manufacturer's standard practices. Exterior color shall be in accordance with color chip 24084 (Green, Semigloss) of FED-STD-595 (see 4.8.4.2).

3.6.3 Serviceability. Servicing, cleaning, and reassembly shall be accomplished without the use of special tools and without disturbing the air cleaner connections to the engine (see 4.8.4.3).

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3.6.4 Interchangeability. The assembly shall be designed and fabricated such that 2 or more items possessing functional and physical characteristics which are equivalent in performance and durability are capable of being exchanged one for the other without alteration of the items themselves or of adjoining items except for adjustment, and without selection for fit, repair parts, or performance (see 4.8.4.4).

3.6.5 Safety. The assembly shall be cleaned of any foreign material which might detract from the intended operation, function, or appearance of the assembly. Screws, nuts, and bolts shall show no evidence of cross threadings, mutilation, or detrimental or hazardous burrs, and shall be firmly secured. Bearing assemblies shall be free of rust, discoloration, and imperfections of ground, honed, or lapped surfaces. Contacting surfaces shall be free of tool marks, gouge marks, nicks or other surface-type defects. There shall be no detrimental interference, binding, or galling (see 4.8.4.5).

3.7 Operating environment requirements.

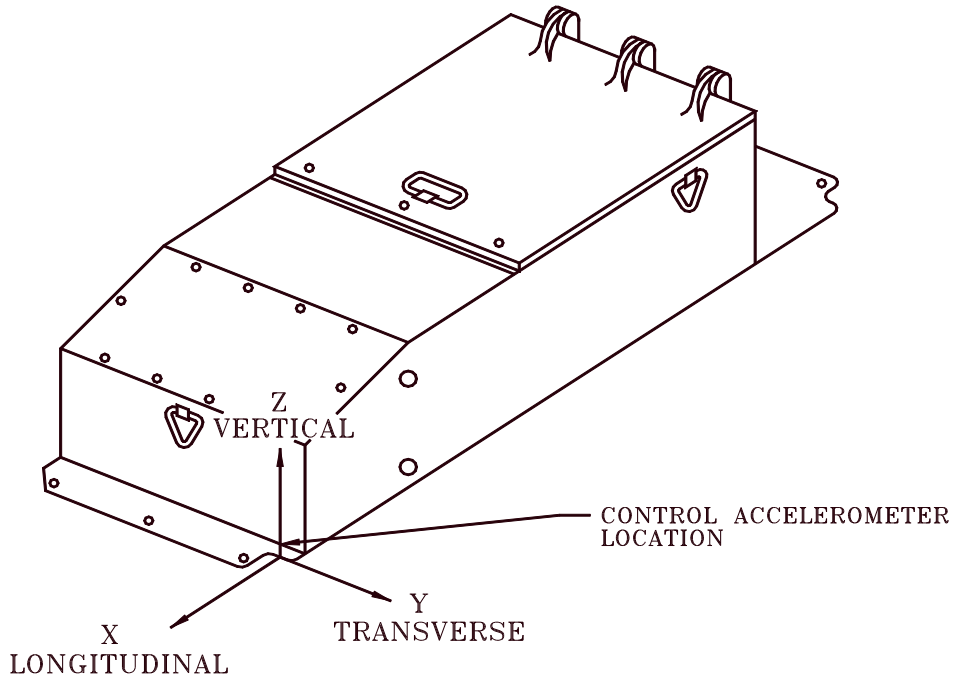
3.7.1 Vibration. The assembly shall be capable of withstanding the vibration profile of figure 1 for a period of 80 minutes (including forced dwells of 13.33 minutes each) in each of the three mutually perpendicular axes (see 4.8.5.1).

3.7.2 Salt fog. The assembly shall withstand exposure to salt fog atmosphere. The salt fog atmosphere shall consist of a salt fog solution defined as 5% by weight NaCl and 95 % by weight distilled water. The exposure zone temperature range shall be 90 to 95ºF. The fog density shall be approximately 3 quarts of solution to each 10 cubic feet (see 4.8.5.2).

4. VERIFICATION

4.1 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision, maintenance, and accuracy of all inspection equipment necessary to assure that supplies and services conform to contract requirements.

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Axis	Sweep acceleration 5 to 500 to 5 Hz at 0.9 octaves/minute	Forced dwells					
	gravity (G)	1st		2nd		3rd	
		Accel. (G-level)	Freq. (Hz)	Accel. (G-level)	Freq. (Hz)	Accel. (G-level)	Freq. (Hz)
Vertical (Z)	0.9	4.2	20	2.4	105	2.4	265
Transverse (Y)	0.9	1.5	50	1.8	100	-	-
Longitudinal (X)	0.9	1.2	245	-	-	-	-

FIGURE 1. Vibration levels for assembly.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
   1. Preproduction inspection (see 4.5.1).
   2. Initial production inspection (see 4.5.2).
b. Conformance inspections (CI) (see 4.6).
   1. Examination (see 4.6.2).
   2. Tests (see 4.6.3).

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4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted in accordance with 4.4 and under the following conditions:

a. Temperature: 73 + 18°F (23 + 10°C).
b. Humidity: 50 + 15%.
c. Density of air: 0.073 lb/ft3 (1.17 kg/m3) at 80°F (27°C).
d. Pressure: 30 + 1 in. of Hg (101 + 3 kPa).

4.4 Inspection procedures. Unless otherwise specified (see 6.2), all test procedures shall be in accordance with 4.3 and the following:

a. The test setup for all tests requiring airflow shall be as shown in figure 2.
b. Dust feed rates shall be averaged over 10-minute intervals.
c. Measured airflow restriction values shall be corrected to the specified air density (see 3.4c) to an accuracy of 0.1 in. of water (25 Pa).
d. Actual airflow values used in tests shall be corrected to standard conditions (see 3.4c) and shall be within + 2% of specified values.
e. The assembly shall be mounted in its normal operating attitude as shown in figure 2 and shall be electrically grounded.
f. In all tests, the assembly shall be connected to a turbine vacuum source as shown in figure 2. The vacuum source shall provide a scavenge airflow as specified in 3.4.2.
g. For first article testing, two filter elements shall be used for each assembly tested. The second filter shall be inserted subsequent to the post service airflow restriction test prior to the dust efficiency test.
h. Air shall be drawn through the assembly with a turbine vacuum source to insure filter element dryness before starting tests.

4.5 First article inspection. First article inspections shall be performed on preproduction or initial production samples as specified in the contract or order.

4.5.1 Preproduction inspection. Unless otherwise specified (see 6.2), the preproduction sample shall consist of two assemblies. Preproduction inspection shall consist of inspection as specified in table II, in the order listed in table III.

4.5.2 Initial production inspection. Unless otherwise specified (see 6.2), the Government shall select two assemblies from the first 10 assemblies, produced under the production contract, for initial production inspection. Initial production units shall be inspected as specified in table II, and in the order listed in table III.

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TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article		CI	
			Preproduction	Initial	Exam	Test
Materials, design and construction	3.2 thru 3.3.2	4.8.1	X	X	X	X
Operating requirements	3.4	4.8.2				X
Initial airflow restriction	3.4.1	4.8.2.1	X	X		X
Dust capacity	3.4.2	4.8.2.2	X	X		
Dust efficiency	3.4.3	4.8.2.3	X	X		X
Post service airflow restriction	3.4.4	4.8.2.4	X	X		
Air leakage resistance	3.4.5	4.8.2.5	X	X		X
Interface requirements	3.5	4.8.3				
Support and ownership requirements	3.6	4.8.4				
Identification marking	3.6.1	4.8.4.1	X	X	X	
Painting	3.6.2	4.8.4.2	X	X	X	
Serviceability	3.6.3	4.8.4.3	X	X		
Interchangeability	3.6.4	4.8.4.4	X	X	X	
Safety	3.6.5	4.8.4.5	X	X	X	
Operating environment requirements	3.7	4.8.5				
Vibration	3.7.1	4.8.5.1	X	X		X
Salt fog	3.7.2	4.8.5.2	X	X		X

TABLE III. Sequence of tests.

Test sequence	Inspection	First article sample	
		1	2
Materials, design, and construction	4.8.1	X	X
Air leakage resistance	4.8.2.5	X	X
Initial airflow restriction	4.8.2.1	X	X
Dust capacity	4.8.2.2	X	X
Post service airflow restriction	4.8.2.4 1/	X	X
Dust efficiency	4.8.2.3	X	X
Vibration	4.8.5.1	X	X

1/ See 4.4 g.

4.6 Conformance inspection. Conformance inspection shall include the examinations of 4.6.2 and the tests of 4.6.3.

4.6.1 Sampling. Sampling shall be as specified in the contract or order (see 6.2).

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4.6.2 Examinations. Samples selected in accordance with 4.6.1 shall be examined as specified in table II, and shall be examined for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Tests. Samples selected in accordance with 4.6.1 shall be subjected to the tests specified in table II. The acceptance number in all cases is zero defects.

4.7 Verification methods. Acceptable verification methods included in this section are visual inspection and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component property analysis, and similarity to previously-approved or previously-qualified designs.

4.7.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3.1).	SIE 1/
103	Identification marking, improper (see 3.6.1).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	SIE
202	Paint, improper application or color (see 3.6.2).	Visual
203	Safety not as specified (see 3.6.5).	Visual

1/ SIE = Standard Inspection Equipment.

4.8 Methods of inspection.

4.8.1 Materials, design and construction. Conformance to 3.2 through 3.3.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data. Examination shall be one or more of the methods outlined in 4.7.

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4.8.2 Operating requirements verification.

4.8.2.1 Initial airflow restriction. To determine conformance to 3.4.1, the assembly shall be subjected to a dust free airflow of 900 ft^3/min (425 L/s) through a new filter element. The restriction to flow shall be measured to verify that the restriction is not greater than 900 ft^3/min (425 L/s). The humidity condition of 3.4 shall not apply.

4.8.2.2 Dust capacity. To determine conformance to 3.4.2, the assembly shall be mounted in the test setup conforming to figure 2, and 1 hour cycles with stepped flow rates as specified in table 1 shall be completed. The restriction to flow (pressure drop through the assembly and absolute filter) shall be measured each time the flow is adjusted to 900 ft^3/min (425 L/sec). Coarse dust shall be fed into the inside of the assembly at 0.025± 0.005 g/ft^3 (0.88± 0.18g/m^3) of actual input airflow. The test shall be conducted until the airflow restriction at 900 ft^3/min (425 L/s) reaches 20 inches of water (5.0 kPa) or 80 hours service life, whichever comes first (see 3.4). Chemical analysis of the coarse test dust shall be in accordance with ISO 5011. Particle size distribution shall be in accordance with ISO 5011. The dust injector to be used will be as specified in ISO 5011. At the end of the test, (1) inspect the clean side of the element and clean side of housing assembly. Any visual signs of dust tracking shall be cause for rejection, (2) verify that a minimum coarse dust accumulative efficiency of 99.9% has been achieved, (3) verify that an 80 hour service life has been achieved at a maximum 20 inches restriction limit at rated air flow and (4) verify that a pre-cleaner efficiency was not less than 97%. The following formulas shall be used to determine coarse dust or pre-cleaner efficiency:

$$\text{Precleaner efficiency (\%)} = \frac{W_1 - [\,(W_4 - W_5) + (W_2 - W_3)\,]}{W_1} \times 100$$

$$\text{Coarse dust efficiency (\%)} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where:

W_1 = Weight of coarse dust fed into the assembly.
W_2 = Weight of absolute filter element and its entrapped dust.
W_3 = Weight of absolute filter element at start of test.
W_4 = Weight of filter element, its entrapped dust and dust encasing the sides of the filter housing compartment.
W_5 = Weight of filter element at start of test.

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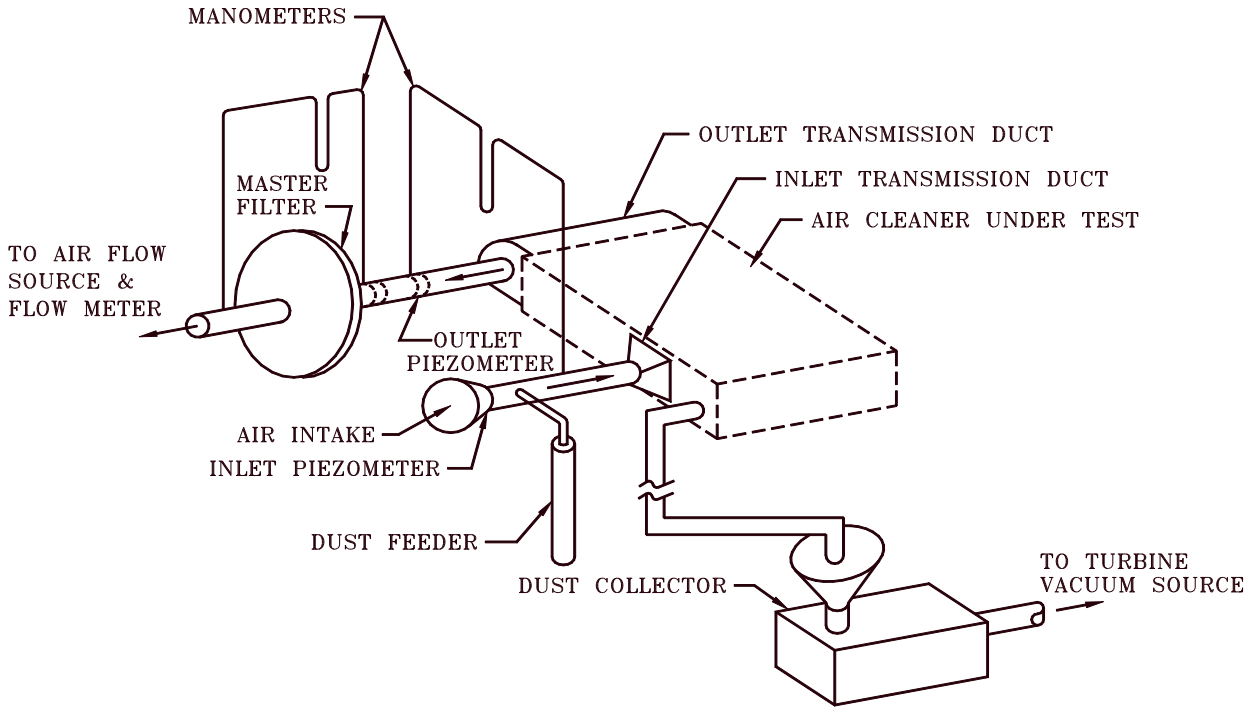


FIGURE 2. Typical test set up (air cleaner with VEDES scavenging).

4.8.2.3 Dust efficiency. To determine conformance to 3.4.3, the assembly shall be mounted in the test setup conforming to figure 2 with an absolute filter (see 4.8.2.3.1) at the output end of the assembly. Fine dust as specified in ISO 5011 shall be placed in a dust feeder and shall be fed into the input side of the assembly at 0.025 $\pm$0.005 g/ft^3 (0.88 $\pm$0.18 g/m^3) of actual input airflow for 30 minutes with an airflow of 900 ft^3/min (425 L/s) and a scavenge airflow of 70 ft^3/min(33 L/s). When the air cleaner has a rated airflow face velocity through the filter element media of 183 cm per minute (6 ft/min) or less, the dust shall be fed for 60 minutes. The dust injector shall be as specified in ISO 5011. Restriction at the start of the test and at 10 minute intervals throughout the test shall be determined. The absolute filter shall be removed from the test setup taking care not to disturb the entrapped dust and reweighed as specified in 4.8.2.3.1. All fine dust and absolute filter element weighings shall be accurate to 0.01 g. Other weighings shall be accurate to 0.01 g wherever possible but never less accurate than 0.1 g. The minimum dust efficiency of the assembly shall be 99.5% when computed using the following formula:

$$\text{Dust Efficiency (\%)} = \frac{W_1 - (W_2 - W_3)}{W_1} \text{ X } 100$$

Where:

W_1 = Weight of dust fed into the assembly.
W_2 = Weight of absolute filter and its entrapped dust.
W_3 = Weight of absolute filter before test.

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4.8.2.3.1 Preparation of the absolute filter. An absolute filter conforming to ASTM C800 shall be placed in an air oven and dried for 6 hours at a temperature of 200°F to 240°F (93°C to 116°C) prior to insertion in the test setup. This drying time may be reduced and the absolute filter considered as completely dry, if there is no change in weight of the absolute filter over two consecutive 10 minute intervals.

4.8.2.4 Post service airflow restriction. To determine conformance to 3.4.4, after the dust capacity test of 4.8.2.2, the filter element shall be removed and cleaned of dust by using clean, dry, compressed air with 100 ±10 lb/in^2 (690 ±69 kPa) line pressure, and a nozzle inside diameter of not less than 0.13 in. (3.18 mm). The air nozzle shall be inserted into each pocket or serration of the filter element using a back and forth motion along the length of each serration and a direct flow into each pocket from inside out. After the filter element is thoroughly air cleaned, it shall be reassembled in the assembly. The assembly shall be subjected to and pass a dust free flow of 900 ft^3/min (425 L/s) of air through the filter element. The restriction to flow shall be measured.

4.8.2.5 Air leakage resistance. To determine conformance to 3.4.5, the assembly shall be sealed at all inlets and outlets and subjected to a vacuum of 30 ±0.5 in. of water (7.5 ±0.12 kPa). The vacuum decay (leakage) shall be monitored for 3 minutes. The humidity condition of 3.4 shall not apply.

4.8.3 Interface requirements verification. To determine conformance to 3.5, the assembly shall be inspected for interface and mounting using one or more of the methods outlined in 4.7.

4.8.4 Support and ownership requirements verification.

4.8.4.1 Identification marking. To determine conformance to 3.6.1, verify the presence of the required markings on the air cleaner assembly. After performing all environmental tests in 4.8.5, re-inspect the assemblies markings for presence and readability.

4.8.4.2 Painting. To determine conformance to 3.6.2, the exterior color shall be examined using one or more of the methods outlined in 4.7 to verify that it is in accordance with color chip 24087 of FED-STD-595.

4.8.4.3 Serviceability. To determine conformance to 3.6.3, the assembly shall be inspected for serviceability using one or more of the methods outlined in 4.7.

4.8.4.4 Interchangeability. To determine conformance to 3.6.4, the interchangeability between different parts, subassemblies, and assemblies having the same military part number shall be verified using one or more of the methods of 4.7.

4.8.4.5 Safety. To determine conformance to 3.6.5, the assembly shall be inspected to assure that the components are free of hazardous burrs, nicks, sharp edges, foreign materials, or

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other imperfections posing physical danger to the operators, and that the assembly and components and securely installed and fastened.

4.8.5 Operating environments requirements verification.

4.8.5.1 Vibration. To determine conformance to 3.7.1, the assembly shall be mounted on vibrating equipment with the filter element inserted and scavenge exhaust attached (see figure 2). The assembly shall vibrate for a period of 80 min. at an amplitude of 0.9 gravity units and a frequency rate from 5 to 500 to 5 Hz in each of the three mutually perpendicular axes as shown in figure 1. The forced dwells as specified in figure 1 shall be performed in place of the resonance search and dwells. Connections and instrumentation shall be attached to permit testing as follows:

a. Test level shall be as specified in figure 3 when conducted at ambient temperatures. The time schedule shall be a total of 80 min. per axis including the forced dwells as specified figure 1.
b. The test item shall be vibrated along each axis in accordance with the condition above. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 4. The specified sweep time is that of an ascending plus a descending sweep and is twice the ascending sweep time shown in figure 4 for the specified range. Linear sweep rates shall be substituted for the logarithmic sweep rate. When linear sweep rates are used, the total frequency range shall be divided into logarithmic frequency bands having similar time intervals such that each time interval is the time of ascending plus a descending sweep for the corresponding band. The sum of these time intervals shall equal the sweep time specified for the applicable frequency range. The linear sweep rate for each band is then determined by dividing each bandwidth in Hz by one half the sweep time in minutes for each band. The logarithmic frequency bands may be readily determined from figure 4. The frequency bands and linear sweep rates shown in table V shall be used for the 2 (or 5) to 500 Hz and 5 to 2000 Hz frequency ranges. For test frequency ranges of 100 Hz or less, no correction of the linear sweep rate is required.
c. After vibration, the assembly shall be examined for structural damage and shall be subjected to an air leakage test as specified in 4.8.2.5 and the dust efficiency test as specified in 4.8.2.3. The filter element shall be cleaned as specified in 4.8.2.4 prior to performing the dust efficiency test.

4.8.5.2 Salt fog. The assembly shall be subjected and inspected to the salt fog requirements of MIL-STD-810 method 509.3 to verify conformance to 3.7.2. After salt fog exposure, the assembly shall be subjected to the efficiency test of 4.8.2.3.

NOTE: A test coupon or witness sample which represents the assembly may be subjected to the salt fog testing and visually inspected in lieu of a full assembly test with prior approval from the procuring activity.

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TABLE V. Linear cycling rates.

Total frequency range (Hz)	Frequency band (Hz)	Sweep time in minutes (min-max-min)	Linear cycling rate (Hz/min)
2 - 500 or	2 to 5	3	2
5 - 500	5 to 22.5	6	5.8
As applicable	22.5 to 110	5	35
	110 to 500	4	195
	5 to 22.5	6	5.8
	22.5 to 110	5	35
5 - 2000	110 to 500	4	195
	500 to 900	3	267
	900 to 2000	2	1100

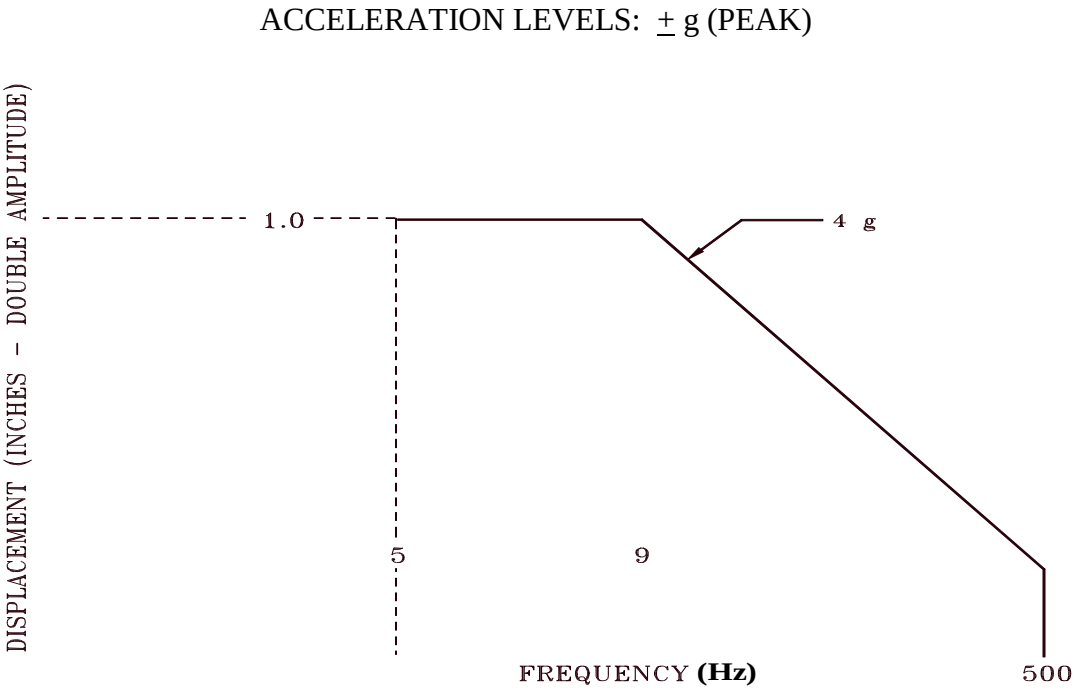


FIGURE 3. Vibration test curve for equipment installed in ground vehicles.

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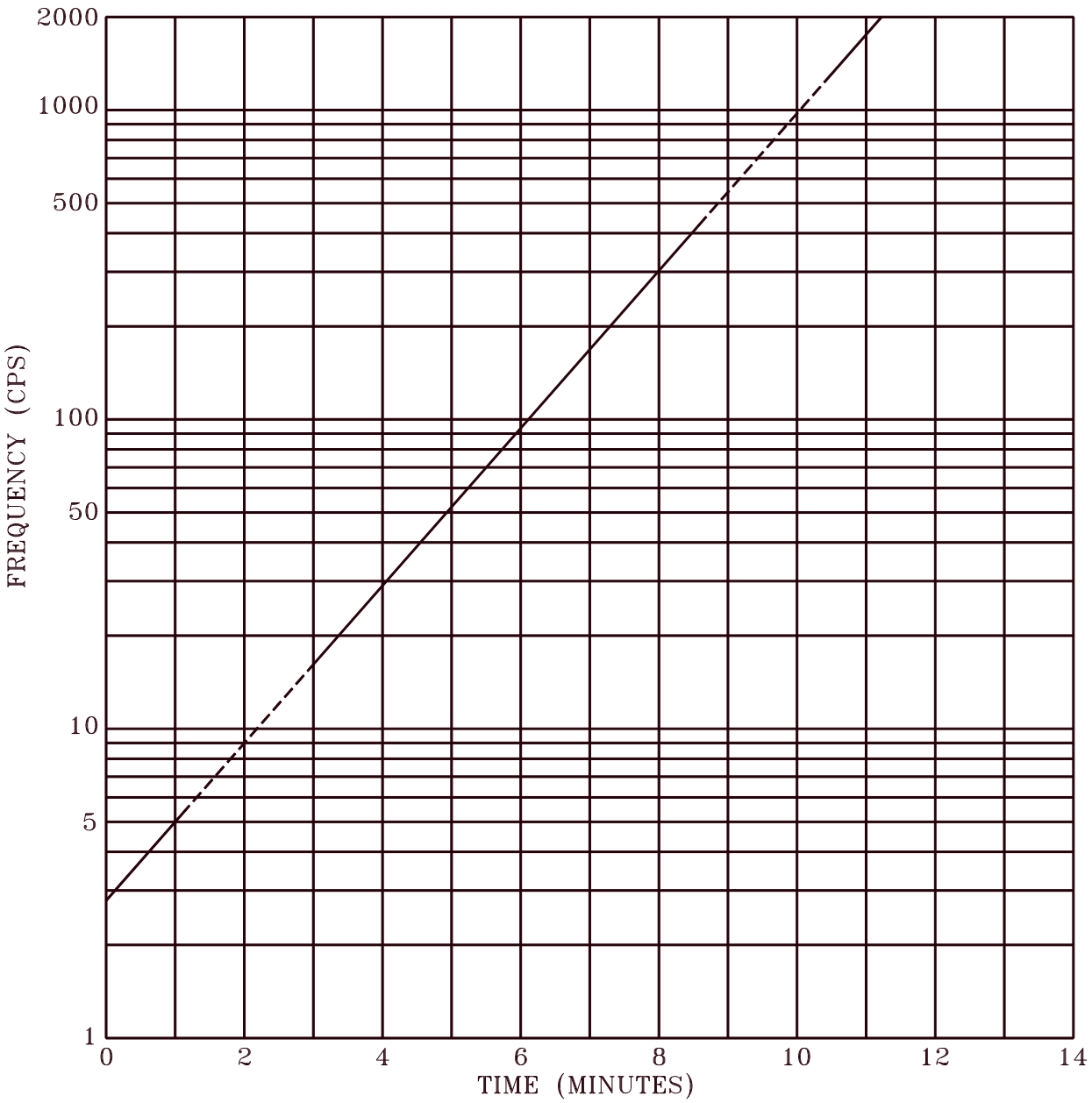


FIGURE 4. Logarithmic sweep.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air cleaner is intended specifically for use on full-tracked, combat M60 series tanks. The air cleaner is military unique because it must operate under extreme dust and vibration combat conditions in all theaters of operation. Commercial air cleaners are not designed to operate under such extreme conditions.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2 and 2.3).
c. When first article inspection is required (see 3.1).
d. Title, number, and date of the applicable drawing (see 3.3).
e. If painting should be other than as specified (see 3.6.2).
f. If responsibility for inspection equipment should be other than as specified (see 4.1).
g. If inspection conditions should be other than as specified (see 4.3).
h. If inspection procedures should be other than as specified (see 4.4).
i. If preproduction sample should be other than as specified (see 4.5.1).
j. If initial production sample should be other than as specified (see 4.5.2).
k. Sampling requirements (see 4.6.1).
l. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Combat vehicle
Dry-type
Dust efficiency
Dust feed
Fender-mounted
Filter element
Flow rate
M60 series tank

6.4 First article inspection. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applies, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table II and their designated frequency based on a risk assessment for the procurement.

6.6 Definitions.

6.6.1 Load capacity. The term used to indicate dust load has caused the air cleaner restriction to reach 20 in. of water (4.98 kPa).

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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